We can measure the actual running time of the execution of an algorithm."
(True / False)

Answer: True

The actual running time of an algorithm can be measured by executing the program on a specific machine with a specific input. However, it depends on hardware and environment, which is why we use Big-O notation for theoretical analysis.

Rearrange the following functions (f1, f2, f3, and f4) in increasing order of asymptotic complexity:

$f1(n) = n^{(3/2)}$, $f2(n) = 2^n$, $f3(n) = n^{\log(n)}$, $f4(n) = n*\log(n)$

This type of question, Follow the below sequence

$$\log n < n < n \log n < n^2 < n^3 < 2^n$$

Why do we ignore the coefficient of the leading term in Time Complexity?

In time complexity analysis, we focus on how an algorithm grows when the input size **n becomes very large**.

In the figure, we see functions like:

- 5n+5
- n^2
- 0.001n^2

Key Observation from the Figure:

1. When **n is small**, a function like 5n+5 may appear larger than 0.001n^2.
2. But when **n becomes very large**, the n^2 function grows much faster than the linear function 5n+5.
3. Although the coefficient is small, a higher power term (like n^2) grows much faster than a lower power term (like n). As n becomes large, the higher power term eventually dominates.
   OR
   Even if the coefficient is very small (like 0.001), the term with the higher power (n^2) will eventually dominate.

Two important properties that an algorithm should have are:

1. Correctness (সঠিকতা)

অ্যালগরিদমটি প্রতিটি বৈধ ইনপুটের জন্য সঠিক আউটপুট দিতে হবে।

অর্থাৎ, সমস্যা যেটা সমাধান করার কথা, সেটি ঠিকভাবে সমাধান করতে হবে।

2. Finiteness (সীমাবদ্ধতা)

অ্যালগরিদম অবশ্যই নির্দিষ্ট সংখ্যক ধাপের পর শেষ হতে হবে।

অর্থাৎ, এটি অসীম লুপে চলতে পারবে না — একসময় থামতেই হবে।

1. Correctness

The algorithm must produce the correct output for every valid input.

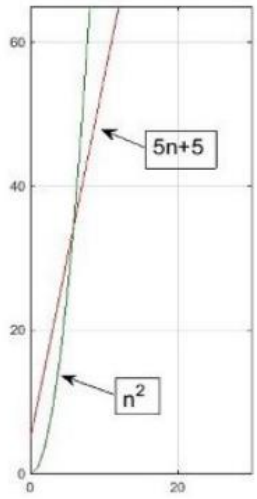
It should properly solve the problem it is designed to solve.

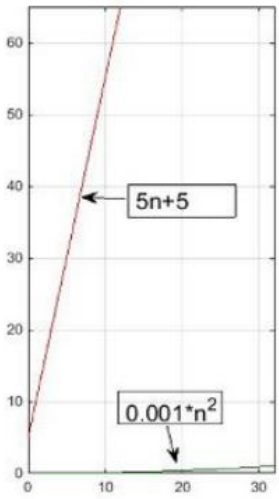
2. Finiteness

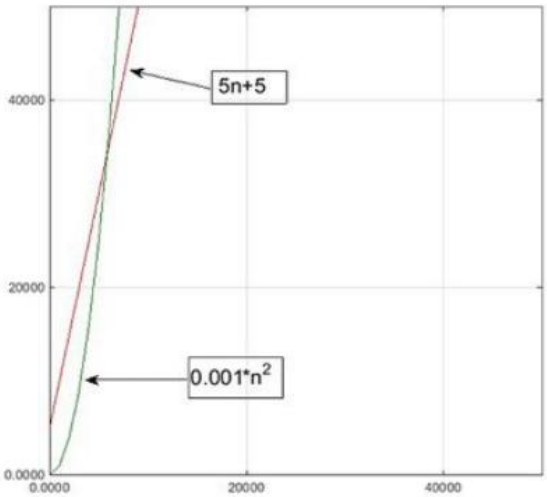
The algorithm must terminate after a finite number of steps.

It should not run forever (no infinite loops).

Insertion sort times complexity:







Asymptotic Notations:
O-notation (Big Oh)

Big-O notation represents the **asymptotic upper bound** of an algorithm.
It describes how the running time grows when the input size (n) becomes very large.

We write:

$$f(n) = O(g(n))$$

This means there exist positive constants c and n_0 such that:

$$0 \leq f(n) \leq cg(n), \quad \text{for all } n \geq n_0$$

That is, for sufficiently large n, the function $f(n)$ does not grow faster than a constant multiple of $g(n)$.

Important Points:

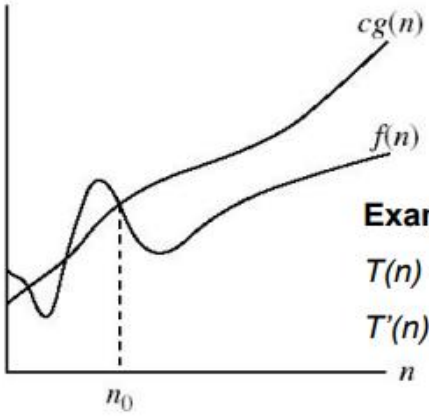
1. Big-O shows the worst-case growth rate.
2. Keep the highest power term.
3. Ignore coefficients.
4. Ignore lower order terms.

Example:
T(n) = 3n^2 + 10n + 8

The highest order term is (n^2).

Therefore,
T(n) = O(n^2)

$O(g(n)) = \{f(n)$: there exist positive constants c and n_0 such that $0 \leq f(n) \leq cg(n)$ for all $n \geq n_0\}$.



O(g(n)) is the set of functions with smaller or same order of growth as g(n)

Examples:

$T(n) = 3n^2+10n\lg n+8$ is $O(n^2)$, $O(n^2\lg n)$, $O(n^3)$, $O(n^4)$, .

$T'(n) = 52n^2+3n^2\lg n+8$ is $O(n^2\lg n)$, $O(n^3)$, $O(n^4)$, ...

Loose upper bounds

$g(n)$ is an ***asymptotic upper bound*** for $f(n)$.

Ω-Notation (Big Omega)

Big-Omega notation represents the **asymptotic lower bound** of an algorithm.
It describes the minimum growth rate of a function for large input size (n).

We write:

$$f(n) = \Omega(g(n))$$

This means there exist positive constants c and n_0 such that:

$$0 \leq cg(n) \leq f(n), \quad \text{for all } n \geq n_0$$

Example:

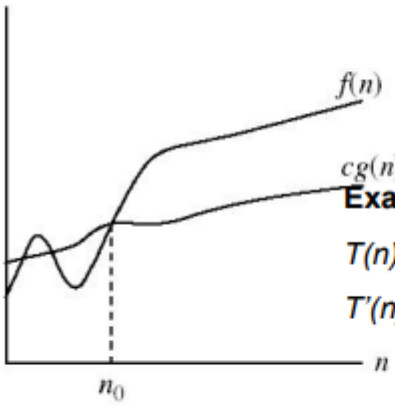
$$T(n) = 3n^2 + 10n + 8$$

Therefore,

$$T(n) = \Omega(n^2)$$

- **Ω - *notation (Big Omega)***

$\Omega(g(n)) = \{f(n)$: there exist positive constants c and n_0 such that $0 \leq cg(n) \leq f(n)$ for all $n \geq n_0\}$.



Ω(g(n)) is the set of functions with larger or same order of growth as g(n)

Examples:

T(n)=3n²+10nlgn+8 is Ω(n²), Ω(nlgn), Ω(n), Ω(lgn),Ω(1)

T'(n) = 52n²+3n²lgn+8 is Ω(n²lgn), Ω(n²), Ω(n), …

$g(n)$ is an ***asymptotic lower bound*** for $f(n)$.

Notation (Big Theta)

Big-Theta notation represents the **asymptotically tight bound** of a function.
It describes the exact growth rate of an algorithm.

We write:

$$f(n) = \Theta(g(n))$$

This means there exist positive constants c_1, c_2, and n_0 such that:

$$0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n), \quad \text{for all } n \geq n_0$$

Example:

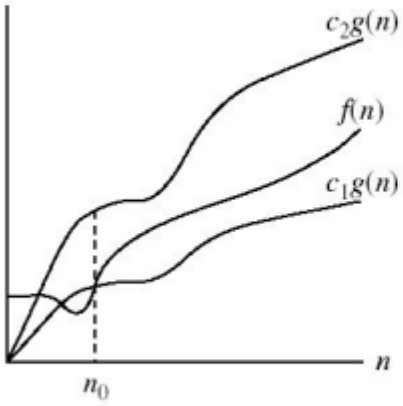
$$T(n) = 3n^2 + 10n + 8$$

Therefore,

$$T(n) = \Theta(n^2)$$

$\Theta(g(n)) = \{f(n)$: there exist positive constants c_1, c_2, and n_0 such that $0 \leq c_1 g(n) \leq f(n) \leq c_2 g(n)$ for all $n \geq n_0\}$.

Θ(g(n)) is the set of functions with the same order of growth as g(n)



* *f(n)* is both *O(g(n))* & *Ω(g(n))* ↔ f(n) is Θ(g(n))

Examples:

T(n) = $3n^2$+10nlgn+8 is Θ(n^2)

T'(n) = $52n^2$+$3n^2$lgn+8 is Θ(n^2lgn)

$g(n)$ is an ***asymptotically tight bound*** for $f(n)$.